

Gamification strategies to improve the motivation and performance in accessibility information collection

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ABSTRACT

The high cost of dispatching professionals to physically audit locations limits the coverage of accessibility maps. Vision-based techniques and crowdsourcing using streetscape imagery are beneficial as they do not entail physical visits by people to actual locations. However, they exhibit limitations such as outdated photos and occlusions. In-the-field crowdsourcing enables physical collection of accessibility information at a low cost. The effectiveness of this technique is contingent on people with free time and high motivation to visit actual locations. In this paper, we introduce a crowdsourcing platform for constructing accessibility maps that support people with diverse free times and motivations. A combination of field auditing and field sensing, coupled with including and excluding gamification is explored. Game types are determined based on the characteristics of field auditing and field sensing. The experimental results confirm that gamification improves the motivation levels and performance of participants in certain aspects of crowdsourced field auditing and crowdsourced field sensing. Furthermore, the results indicate that item collection games have a greater impact on the motivation for taking unusual roads to collect accessibility information than continuous walking games.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility systems and tools.**

KEYWORDS

accessibility map, crowdsourcing, deep learning, gamification

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1 INTRODUCTION

Accessibility maps facilitate safe and facile movement of people with mobility problems. Conventionally, these maps were created through manual audits conducted by professional staff from public facilities. However, the high costs incurred when dispatching professional auditors to actual locations limits the coverage of accessibility maps [1, 13, 44]. Two methods have been proposed to eliminate the need for auditors to physically examine actual locations. The first method, *automatic auditing* [3, 24, 48, 49], utilizes streetscape imagery coupled with computer vision techniques to detect accessibility problems. In contrast, the second method, *crowdsourced virtual auditing* [5, 7, 10, 13, 18, 19, 22, 23, 29, 44–46, 51], relies on online crowdworkers to remotely audit accessibility problems by virtually walking through streets using street view services. Both methods are advantageous in that, they do not require auditors to physically visit actual locations. However, they also manifest limitations such as outdated photos, occlusions, and sparseness in traffic-free zones [1, 7, 10, 18, 29, 45, 49, 51]. As these limitations adversely affect accessibility, physical audits are thus far, necessary. To collect accessibility information from actual locations at a low cost, two types of crowdsourcing methods involving physical audits were proposed. *Crowdsourced field auditing* [9, 11, 15, 33–36, 39, 42, 43] entails physical visits by crowdworkers to actual locations to report accessibility problems. *Crowdsourced field sensing* [9, 12, 16, 26, 27, 30–32, 37, 42, 50] enables gathering sensor data from actual locations in the field, envisaging crowdworkers as sensor nodes in the field. Both methods exhibit the advantages of being able to obtain accessibility information or sensor data from actual locations at a low cost. However, the effectiveness of both methods are contingent on, the motivation level and cooperation of

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people with free time. Consequently, a crowdsourcing method capable of supporting people with diverse free times and motivations [38] is vital in creating accessibility maps.

In this study, we have focused on techniques to increase motivation and user engagement when gathering accessibility information. Specifically, we leverage the power of gamification. Gamification is defined as *the use of game design elements in non-game contexts* [14], and has been widely adopted in domains such as education, work, and well-being [21, 28, 47]. Gamification is also utilized in crowdsourcing [4, 6, 8, 25, 40, 41]. However, with the exception of a few studies [43, 50], the effects of gamification as applied to crowdsourced accessibility information collection have rarely been reported. To expound the effects, we present two sets of two-week experiments that evaluate the impact of game elements on motivation and performance in the contexts of crowdsourced field auditing and crowdsourced field sensing.

2 RELATED WORK

Several methods have been proposed to collect accessibility information. We divided these into the following four groups: automatic auditing, crowdsourced virtual auditing, crowdsourced field auditing, and crowdsourced field sensing.

2.1 Automatic auditing

Several researchers have attempted to automatically detect accessibility problems by applying computer vision techniques to satellite/streetscape imagery. Sun et al. [48] proposed a Siamese neural network-based model to detect missing objects such as curb ramps in streetscape imagery. However, this study employed a small training dataset and focused only on curb ramps. Weld et al. [49] utilized a larger dataset to train a ResNet-based model. This model detected missing curb ramps, obstructions, and surface problems of sidewalks from streetscape imagery. Some studies have combined automated auditing with human validation. Hara et al. [24] proposed a system that automatically detected and superimposed curb ramps on streetscape imagery. This system provided a function for crowdsourced workers to eliminate misidentified curb ramps. Ahmetovic et al. [2] proposed a cascaded computer vision technique for extracting crosswalks from satellite imagery. These were then validated using streetscape imagery. In a subsequent study, they introduced a crowdsourced human validation step to improve the quality of the automatic detection results [3].

The aforementioned methods have the advantages of cost and time efficiency [1, 45]. Despite this, a major drawback is the time lapse between recording the images and the present condition [1, 45]. Developments in rural areas and redevelopment in urban areas can frequently alter the conditions of crosswalks and curb ramps. Furthermore, diverse studies have revealed the indiscernibility of certain features on sidewalks due to occlusions caused by roadside objects, such as parked cars and trees [1, 45, 49].

2.2 Crowdsourced virtual auditing

Several studies have explored the remote assessment of street characteristics using geospatial collections. Some studies developed Google Street View based tools for auditing neighborhood environments [5, 7, 18]. These tools allow users to remotely assess sidewalk

characteristics such as the existence of curb ramps, surface quality, lighting, and safety. To improve the quality of existing audit data, an online worker filtering mechanism integrated with a verification interface was developed [23]. Some researchers have also investigated the factors that affect the performance of crowdsourced virtual auditing. Clarke et al. [13], Rundle et al. [44], and Gullón et al. [19] compared the performance of crowdsourced virtual auditing and field auditing. Zhu et al. [51] investigated the performance of online workers possessing varying degrees of familiarity of a region. Saha et al. [46] compared the auditing quality between paid and unpaid crowdworkers. Hanibuchi et al. [22] examined the performance of auditors categorized according to age, gender, and residence location. These studies indicated that the performance of crowdsourced virtual auditing depends on a variety of factors.

Similar to automatic audits, crowdsourced virtual audits are advantageous in terms of cost and time efficiency. They also exhibit drawbacks due to time lapse and occlusions [1, 10, 45]. Several studies have reported objects such as buses, parked cars, road works, and trees hiding sidewalks [7, 18, 29, 51]. Gullón et al. [19] reported that virtual audits were longer than field audits for complex streets in continental European cities. In addition, we confirmed that streetscape images were frequently unavailable for streets of traffic-free zones where image-capturing cars could not enter.

2.3 Crowdsourced field auditing

Geographic information provided by untrained citizens is termed as volunteered geographic information (VGI). The availability of the global positioning system (GPS), smartphones, and map resources has augmented the collection and utilization of VGI [17, 20]. A typical example is a citizen reporting tool for non-emergency urban issues. This encompasses many platforms such as SeeClickFix and FixMyStreet, which permit citizens to report a free text description of urban issues [33, 36]. Cila et al. [11] suggested a system design to enhance citizen engagement by proposing badges as rewards. Existing geographic information services namely, Google Maps and FourSquare, facilitate information collection by allowing users to report accessibility information such as wheelchair accessibility on specific points of interest [15]. Wheelmap [39] is a dedicated platform with a simple rating interface for reporting wheelchair accessibility of places by crowdsourcing. A few other studies have combined manually reported accessibility data with heterogeneous data such as sensor data (see Section 2.4) [9, 42], public transportation data [34], and authoritative data sources [42].

The aforementioned methods are advantageous in that, they utilize accessibility information reported by citizens from actual locations. These data are not affected by outdated photos, occlusions, and sparseness in traffic-free zones. Data collection can be performed at a lower cost than that expended by local administrations. Nevertheless, one drawback of this method is that, it is impractical to provide continuous incentives to several auditors in order to physically visit locations. Menkens et al. [35] introduced a reputation-based mechanism to motivate users to gather information on wheelchair accessibility. Prandi et al. [43] and Yamato et al. [50] leveraged the power of gamification by enabling users to collect accessibility information by playing games.

2.4 Crowdsourced field sensing

Crowdsourced field sensing and crowdsourced field auditing are similar in that, data are uploaded by in-the-field auditors. However, they differ in terms of task explicitness and workload. In crowdsourced field sensing, sensors automatically record data. This eliminates the need for users to manually report problems. Typically, inertial sensors attached to wheelchairs recorded the data. Accessibility problems such as curbs, steps, and slopes were detected from the sensor data using machine learning techniques [12, 27, 32]. To record details of problematic locations, other studies introduced cameras that captured sidewalk surfaces when inertial sensors detected large movements of wheelchairs [16, 30]. Isezaki et al. [26] studied the psychological and physical aspects of accessibility. They used a combination of heartbeat sensor and inertial sensors to study the comfort when traversing problematic sidewalks in a wheelchair.

The advantages and drawbacks of crowdsourced field sensing are similar to those of crowdsourced field auditing. These approaches are unaffected by problems arising from car-captured imagery. However, they do not motivate users. Furthermore, this approach of attaching sensors to wheelchairs is limited by the small number of wheelchair users, leading to poor coverage of the area. To overcome this problem, a few studies have acquired sensor data from smartphones of pedestrians [37, 42, 50]. In accordance with this approach, Kurauchi et al. [31] proposed a method to detect accessibility problems by uniformly analyzing sensor data generated from users by employing different modes of travel such as walking, wheelchair, stroller, and bicycle.

3 SYSTEM DESIGN

This study focuses on crowdsourced field auditing and crowdsourced field sensing because both methods are devoid of the drawbacks of streetscape imagery, namely outdated photos, occlusions, and sparseness in traffic-free zones. Additionally, they enable the collection of accessibility information or sensor data from actual locations at a low cost. To avoid excessive dependence on individual auditor attributes (especially, having free times and high motivations), our platform adopts a system design that supports people with diverse free times and motivation levels as indicated in Fig. 1 (a), in accordance with our previous study [38]. This platform provides a smartphone application with four modes: *Reporter*, *Gaming reporter*, *Walker*, and *Gaming walker*, as depicted in Fig. 1 (b).

Our study comprises two main data paths to gather accessibility information. The first data path applies to data judged by a human (the left half of Fig. 1 (a)). Because manual reporting can be time-consuming, this path is appropriate for participants with plenty of time on hand. The second data path applies to AI-judged data (the right half of Fig. 1 (a)). As it is devoid of detailed manual reporting, this path is suited to relatively busy participants. Each data path is equipped with a gamification mode (the bottom of Fig. 1 (a)) to encourage less motivated participants. This design allows participants to select a convenient method depending on their free time and motivation level.

3.1 Reporter

This mode pertains to participants with plenty of free time and high motivation. In this mode, participants manually report accessibility

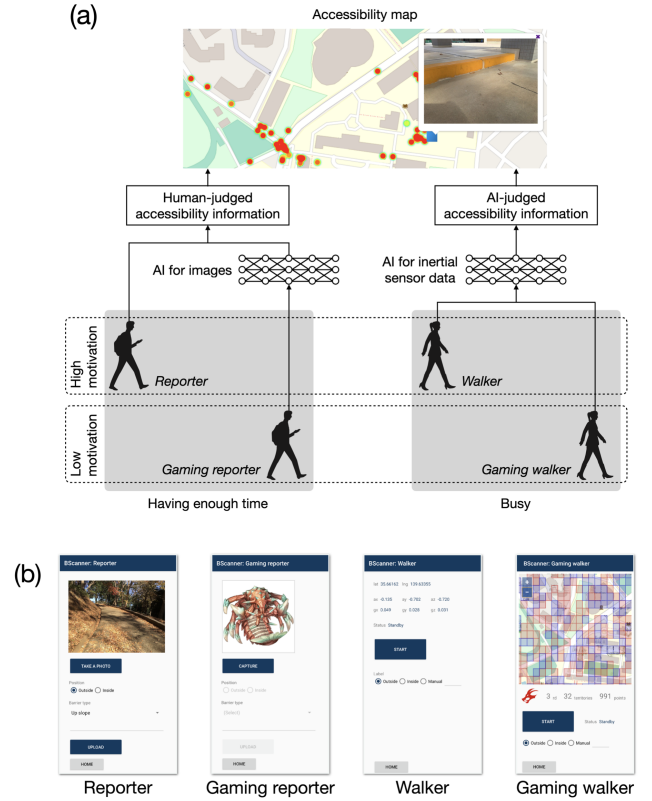


Figure 1: (a) System design of proposed platform. (b) Appearance of smartphone client in each mode.

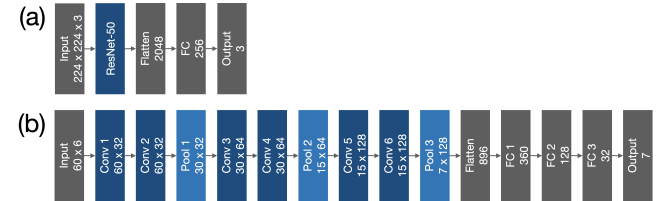


Figure 2: (a) ResNet-50 based Neural Network for images. (b) 1D convolutional neural network for inertial sensor data. FC represents fully connected layer, Conv represents convolutional layer, and Pool represents pooling layer.

problems using photos, descriptions, and location information. The participants employing this mode act as follows. First, they audit sidewalks and find an accessibility problem. Second, they take a photo of the problem. Third, they select an appropriate label for the problem such as stairs, from a list of options. Finally, they upload the photo and the label along with the location information measured by the GPS unit of a smartphone.

3.2 Gaming reporter

This mode provides the gamification mode of *Reporter* for participants with plenty of free time but with less motivation. Accessibility

problems are manually reported by participants by sharing photos and location information. Considering the compatibility with the act of taking photos, we introduced an item collection type of game. Specifically, in this game, participants collect *monsters* by taking photos of accessibility problems. Each monster corresponds to an accessibility problem, e.g., a monster associated with stairs has a stairs-shaped part on its body. In addition to this, two types of monsters, *normal* and *rare*, are prepared for each type of accessibility problem. *Rare* monsters are acquired when users report an accessibility problem of a type that has not been reported nearby. From the photo, an image analyzer (*AI for images* in Fig. 1 (a)), developed using the ResNet-50 based neural network shown in Fig. 2 (a), estimates the type of accessibility problem, namely *step*, *stairs*, or *slope*, with an F-measure of 0.95. The participants employing this mode act as follows. First, they audit sidewalks and identify a potential accessibility problem. Second, they take photos of the problem. Third, they upload the photos with location information tagged using the built-in GPS of smartphone. Finally, they obtain a monster corresponding to the reported accessibility problem.

3.3 Walker

In this mode, the participants are those who lack sufficient free time but possess high motivation. Accessibility problems are detected by analyzing sensor data obtained from smartphones of walkers. Participants are not required to manually detect or describe accessibility problems. The participants employing this mode act as follows. First, they launch the application in this mode. Second, they place the smartphone in their trouser pocket. Finally, they walk for their own purposes (e.g., commuting). At this time, the application records the acceleration, angular velocity, and location information, such as latitude and longitude, at a frequency of 20 Hz; these variables are measured by the smartphone's accelerometer, gyroscope, and GPS unit, respectively. Our system analyzes the data in the following sequence. First, the system fragments all data into small bits such that, the length of each fragment is equivalent to 3 s of walking. Specifically, each fragment of data comprises six sets of triaxial acceleration and triaxial angular velocity recorded at 20 Hz for 3 s, cumulatively leading to 60 samples. Second, the system analyzes each data fragment and estimates the type of road by using the 1D convolutional neural network (Fig. 2 (b)) to detect accessibility problems. This network is capable of estimating seven road types, namely *flat*, *up-step*, *down-step*, *up-stairs*, *down-stairs*, *up-high slope*, and *down-high slope*, with an F-measure of 0.99. Finally, the system associates each estimated road type with location information.

3.4 Gaming walker

This mode provides the gamification mode of *Walker* for participants lacking adequate free time and possessing low motivation. In this mode, accessibility problems are detected by analyzing sensor data obtained from smartphones in the same manner as that of *Walker*. To make the act of walking itself the main element of the game, we introduce a continuous walking type of game. In this game, players are divided into two teams (blue vs red) where they compete in obtaining a larger area of territory by walking. This

game is designed based on prior knowledge that inter-team competition leads to higher enjoyment and motivation among individuals to participate in crowdsourcing tasks [41]. In case participants choose to run in lieu of walking, they cannot expand their territories. These safeguards in the game are necessary as running reduces the estimation performance of the neural network. The participants employing this mode act as follows. First, they find a virtual area to be obtained using the game map that shows the territory of the blue and red teams. Second, they go to the real area. Third, they place their smartphone in their trouser pocket. Finally, they walk to obtain the virtual area. Simultaneously, the application records the data in the same format as that recorded by *Walker*.

4 PRELIMINARY EXPERIMENTS

Several studies have demonstrated that gamification can improve motivation and performance in crowdsourcing tasks [4, 6, 8, 25, 40, 41]. However, a handful of studies have explored the effects of gamification on crowdsourced accessibility information collection [43, 50]. Hence, we formulated the following research questions and conducted two preliminary experiments to answer these questions.

RQ1. *Does gamification improve the motivation and performance of participants in crowdsourced field auditing?*

RQ2. *Does gamification improve the motivation and performance of participants in crowdsourced field sensing?*

To evaluate the motivation of the participants, the following questionnaire was used in both experiments 1 and 2. The participants answered these questions using a seven-point Likert scale ranging from 1 (disagree) to 7 (agree).

Q1. *Did you take any unusual roads using this mode?*

Q2. *Do you want to continue collecting accessibility information using this mode?*

4.1 Experiment 1: Crowdsourced field auditing

4.1.1 Purpose and participants. The purpose of this experiment was to answer RQ1. Specifically, we compared the motivation and performance of participants between two crowdsourced field auditing modes; *Reporter* (without gamification) and *Gaming reporter* (with gamification). This experiment was conducted using a within-subject design. Eight university students having no disabilities participated in this experiment without compensation.

4.1.2 Procedure. The experimental period was divided into the first and second periods. In the first period, each participant used the application for one week. The mode, namely *Reporter* or *Gaming reporter*, was randomly selected. This was followed by answering the questionnaire on the last day of the period. Subsequently, each participant used the application for one week in the alternative mode, followed by answering the questionnaire on the last day of the second period. In this experiment, both modes provided participants with rankings based on the number of data/monsters collected. To compensate for the small number of participants, we introduced computer-controlled virtual participants that collected a certain number of data/monsters each day.

4.1.3 Results and discussion. In terms of motivation of participants, Fig. 3 (a) shows the distribution of responses to the questionnaire. The number of participants who marked 5 and over for both Q1 and

Q2 was higher in *Gaming reporter* than in *Reporter*. The Wilcoxon signed-rank test at a significance level of 0.05 showed that both Q1 and Q2 exhibited significant differences between *Reporter* and *Gaming reporter*. This was endorsed by comments from most participants such as “I was looking forward to seeing what kind of monsters I could acquire.” The provision to collect *normal* or *rare* monsters for *steps*, *stairs*, and *slopes* was found to contribute to an increase in motivation levels in *Gaming reporter*. Furthermore, participants were unaware of the type of monsters that could be acquired without going to actual locations. The absence of preliminary clues also instigated participants to traverse unexplored routes.

Comparing the performance of both modes as evidenced from Fig. 3 (b), no significant difference was observed between the numbers of images posted using each mode. Most participants commented “I enjoyed the game, but it wasn’t enough to go out.” This indicates that our game is deficient in its intensity. Collaboration with professional game creators can be explored to solve this problem. Furthermore, the decrease in opportunities to venture outside due to the pandemic may have contributed to this adverse response from participants.

Based on the outcomes of the studies above, we answer RQ1 as follows: Games that provide the experience of item collection, such as our monster collecting game and *Pokémon GO*, potentially contribute towards improving the motivation for crowdsourced field auditing. However, the impact of gamification on the performance of participants is unclear owing to insufficient experimental data.

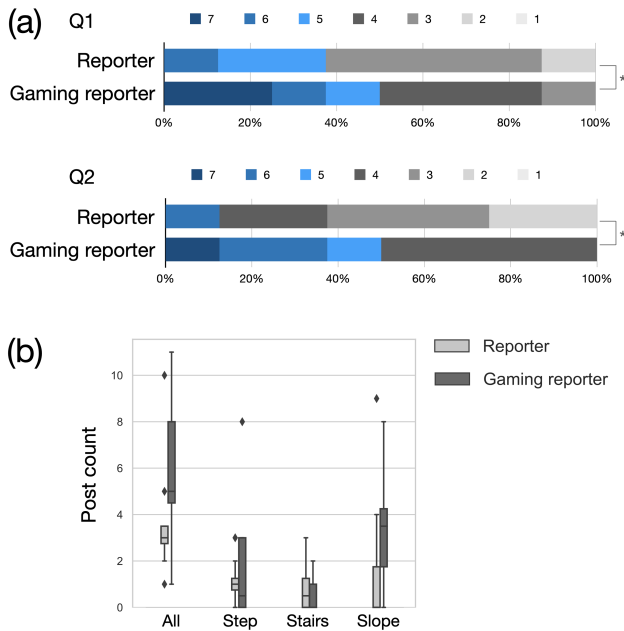


Figure 3: (a) Distribution of responses to Q1 and Q2 for Experiment 1 (N = 8). (b) Post counts of images (N = 8).

4.2 Experiment 2: Crowdsourced field sensing

4.2.1 Purpose and participants. The purpose of this experiment was to determine RQ2. Specifically, we compared the motivation and performance of participants between two crowdsourced field

sensing modes, namely *Walker* (without gamification) and *Gaming walker* (with gamification). This experiment was conducted by utilizing a within-subject design with eight university students who received no compensation. None of the students had any disabilities. Additionally, some of the participants from Experiment 1 were retained here.

4.2.2 Procedure. The procedure for this experiment was similar to that of Experiment 1, consisting of two periods. In the first period, each participant used the application for one week in the randomly determined *Walker* or *Gaming walker* mode. The questionnaire was completed on the last day of the period. During the second period of one week, each participant used the application in the other mode, and then responded to the questionnaire on the final day. Competitors were necessary in *Gaming reporter*; to compensate for the small number of participants, we introduced computer-controlled virtual participants who garnered territories around each participant at a certain pace.

4.2.3 Results and discussion. Fig. 4 (a) shows the distribution of responses to the questionnaire signifying the motivation levels of participants. The number of participants who scored 5 and over for both Q1 and Q2 was higher in the case of *Gaming walker* than *Walker*. However, the Wilcoxon signed-rank test at a significance level of 0.05 revealed that only Q2 exhibited a significant difference between *Walker* and *Gaming walker*. Two possible reasons can be attributed to this finding. First, unlike *Gaming reporter*, participants were required to continuously walk a certain distance to achieve goals in *Gaming walker*. This may have resulted in the participants taking *usual* roads. Comments from some participants such as “It was easier to play on familiar streets.” supported this proposition. Second, participants could easily predict the outcome. This means that participants were certain that they would obtain the territories that they traversed. This could have potentially decreased the uncertainty of the game, thereby reducing the incentive for participants to traverse unusual roads. A possible solution is to introduce a large number of virtual participants. Thus, the shortage of obtainable areas on usual roads would necessitate physical participants to expand their walking areas to win the competition and garner new areas.

In terms of performance, the Wilcoxon signed-rank test¹ at a significance level of 0.001 indicated a significant difference between the distances walked by participants using each mode, as evidenced from Fig. 4 (b). Most of the participants commented that they walked long distances as they were motivated by the visual representation of the territory expanding as they traversed it.

From the outcomes of the above studies, we answer RQ2 as follows: Games imparting the experience of continuous walking, such as our territory expanding game and *Ingress*, potentially improve the motivation and performance in crowdsourced field sensing. However, if the number of participants is limited, there is little impact on the motivation for taking unusual roads.

¹We used the Wilcoxon signed-rank test instead of the t-test because the Shapiro-Wilk test did not confirm the normality of the data.

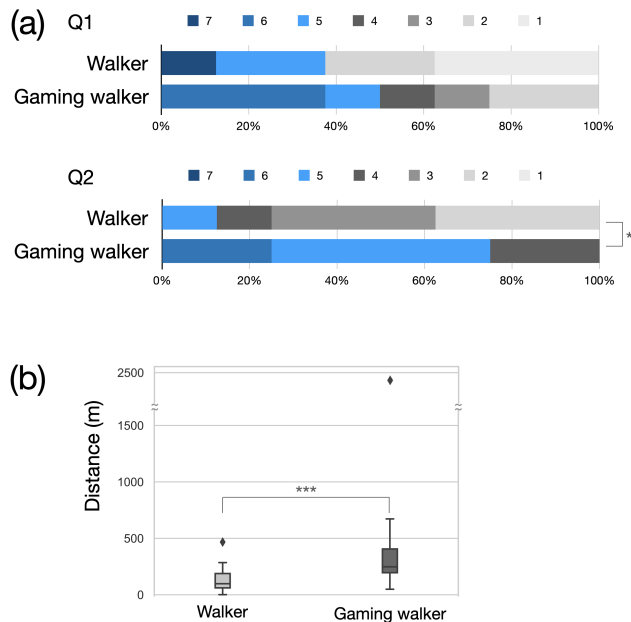


Figure 4: (a) Distribution of responses to Q1 and Q2 for Experiment 2 (N = 8). (b) Walked distances using each mode (N = 8).

5 CONCLUSION AND FUTURE WORK

This paper details our crowdsourcing platform for constructing accessibility maps. We employed a combination of field auditing and field sensing for people with diverse free time to participate. Moreover, a combination of with and without gamification was employed for people with different motivations to participate. Specifically, considering the differences in the explicitness and workload of each task, we applied the item collection type of game for field auditing and the continuous walking type of game for field sensing, respectively. The results of preliminary experiments confirmed that gamification improves the motivation and performance of participants in certain aspects of crowdsourced field auditing/sensing. Furthermore, the results indicated that item collection games have a greater impact on the motivation for taking unusual roads than continuous walking games. These findings can contribute to activities for accessibility information collection with high coverage. The main limitations of this study are as follows. First, the numbers of participants in the experiments were small. To complement the shortage of numbers, virtual participants were introduced to make the real participants competitive. Second, although unpaid, the participants used the platform at the request of the experimenters. Only one request was made at the beginning of the experiment, however, this may have affected their motivations. Third, it has not been verified whether the accessibility problems reported by the participants were beneficial to people with mobility difficulties. To overcome these limitations, we are currently preparing for a two-month field experiment with 50+ participants including people with mobility difficulties.

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